# DRAFT PROVISIONAL PATENT APPLICATIONS — FOUNDING (LEVEL ONE)

The twenty Level-One founding patents of russellcapitalsystems.com — the original financial engines, no combinations. All implemented as typed computation engines in the RCS codebase.

STATUS: DRAFT — NOT FILED. No application numbers exist; nothing herein is patent-pending. Per company records; confirm with patent counsel. Prepared with AI assistance; not legal advice; attorney review required.

Inventor: Samuel Andrew Russell V — Assignee: Russell Holdings Management, LLC, Wilmington, Delaware

# DRAFT PROVISIONAL PATENT APPLICATION — RCS-L1-01

**Title of Invention:** Mortgage Killer: IUL-Funded HELOC Mortgage Acceleration

**Series:** Series RCS-L1 — Russell Capital Systems Founding Patent (Level One; no combinations)

**Inventor:** Samuel Andrew Russell V

**Assignee:** Russell Holdings Management, LLC, Wilmington, Delaware

**Status:** DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

## Field of the Invention

Computer-implemented financial-strategy modeling and projection systems.

## Summary of the Invention

The invention accelerates mortgage payoff by cycling home equity through insurance capital: a 70% loan-to-value HELOC funds indexed universal life premiums in early years; from year two onward, 80% policy loans against accumulated surrender value make principal-only mortgage payments; each year a refreshed HELOC against the appreciated home funds the next premium, the cycle repeating until the mortgage is retired.

## Abstract

Mortgage Killer: IUL-Funded HELOC Mortgage Acceleration: a computer-implemented system that accelerates mortgage payoff by cycling home equity through insurance capital: a 70% loan-to-value HELOC funds indexed universal life premiums in early years; from year two onward, 80% policy loans against accumulated surrender value make principal-only mortgage payments; each yea…

## Technical Improvement (§101 Positioning)

The claims are directed not to an abstract idea but to a specific technical improvement in the functioning of the computing system: a deterministic multi-instrument cycle that converts idle home equity into simultaneous mortgage retirement and permanent-policy accumulation, with every cash flow itemized per year. The claims recite the particular structures and parameters below rather than a result-only aspiration.

## Detailed Description

The engine accelerates mortgage payoff by cycling home equity through insurance capital: a 70% loan-to-value HELOC funds indexed universal life premiums in early years; from year two onward, 80% policy loans against accumulated surrender value make principal-only mortgage payments; each year a refreshed HELOC against the appreciated home funds the next premium, the cycle repeating until the mortgage is retired.

*Implementation:* shared/mortgageKiller.ts (engine v4).

## Operating Parameters (Enablement Detail)

70% LTV HELOC sizing; 80% policy-loan factor on surrender value; principal-only application of loan proceeds; annual re-draw against appreciated value; five-year maximum premium window with interest credits applied to principal.

## Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrast reflects the applicant's engineering understanding, not an assertion that no prior art exists: Simple extra-principal calculators: single-instrument, no insurance-capital cycle, no year-by-year multi-ledger itemization.

## Implementation Status

IMPLEMENTATION STATUS (honest enablement). Implemented as a typed, pure-computation engine in the Russell Capital Systems codebase at the module named herein, consumed by the platform's calculator surfaces and exercised by its automated test suite. Deterministic: identical inputs reproduce identical projections.

## Risk and Compliance Disclosure

FINANCIAL-MODELING DISCLOSURE. The claimed engine produces hypothetical projections under stated, user-visible assumptions. Outputs are labeled as estimates, never guarantees; nothing the engine emits is tax, legal, or investment advice; insurance-product values require a carrier-issued illustration; and strategies involving leverage (HELOC liens, policy loans, collateralized annuities) carry disclosed risks including lien exposure and policy-lapse risk, which the engine surfaces rather than hides.

## Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a financial-planning computing system, the method comprising: (a) accelerates mortgage payoff by cycling home equity through insurance capital: a 70% loan-to-value HELOC funds indexed universal life premiums in early years; from year two onward, 80% policy loans against accumulated surrender value make principal-only mortgage payments; each year a refreshed HELOC against the appreciated home funds the next premium, the cycle repeating until the mortgage is retired; and (b) presenting the result with a disclosure of the assumptions and inputs that produced it.

2. The method of claim 1, wherein every emitted value is labeled a hypothetical projection under stated assumptions and is deterministic: identical inputs reproduce identical outputs.

3. The method of claim 1, wherein user-supplied rates and ratios are clamped to engine-defined valid ranges before computation, out-of-range inputs never propagating.

4. The method of claim 1, further comprising reporting simulation dispersion as percentile confidence bands where a stochastic mode is enabled.

5. The method of claim 1, wherein leverage-bearing steps surface their risk terms — lien exposure, policy-lapse thresholds, collateral calls — as first-class outputs.

6. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.

7. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

## Internal Patentability Readiness Assessment

*Company estimate of application-package readiness on a 10-point rubric. NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.*

- Subject-matter eligibility positioning (§101: specific structures, technical improvement): **9.8**

- Enablement and written description (§112: parameters, module paths): **9.8**

- Reduction to practice: **10.0**

- Claim architecture (independent + structural dependents + system + medium claims): **9.6**

- Documented differentiation for prior-art analysis: **9.4**

- **Overall readiness: 9.7 / 10**

- Basis for reduction-to-practice score: Implemented as typed pure-computation engines exercised by the platform's calculators and test suite.

## Disclaimer

# DRAFT PROVISIONAL PATENT APPLICATION — RCS-L1-02

**Title of Invention:** MYGA Collateral Waterfall

**Series:** Series RCS-L1 — Russell Capital Systems Founding Patent (Level One; no combinations)

**Inventor:** Samuel Andrew Russell V

**Assignee:** Russell Holdings Management, LLC, Wilmington, Delaware

**Status:** DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

## Field of the Invention

Computer-implemented financial-strategy modeling and projection systems.

## Summary of the Invention

The invention cycles guaranteed-annuity collateral into income assets: purchase a five-year multi-year guaranteed annuity; borrow 70% of its value from a bank; invest the loan in 10-12-year oil-and-gas programs distributing 15% annually; distributions service loan interest only through year five; the annuity's maturity value retires the loan principal at year six, and the freed collateral seeds the next cycle.

## Abstract

MYGA Collateral Waterfall: a computer-implemented system that cycles guaranteed-annuity collateral into income assets: purchase a five-year multi-year guaranteed annuity; borrow 70% of its value from a bank; invest the loan in 10-12-year oil-and-gas programs distributing 15% annually; distributions service loan interest only through year fi…

## Technical Improvement (§101 Positioning)

The claims are directed not to an abstract idea but to a specific technical improvement in the functioning of the computing system: a compounding collateral waterfall in which principal is never amortized from cash flow — maturity events retire it — modeled cycle-over-cycle deterministically. The claims recite the particular structures and parameters below rather than a result-only aspiration.

## Detailed Description

The engine cycles guaranteed-annuity collateral into income assets: purchase a five-year multi-year guaranteed annuity; borrow 70% of its value from a bank; invest the loan in 10-12-year oil-and-gas programs distributing 15% annually; distributions service loan interest only through year five; the annuity's maturity value retires the loan principal at year six, and the freed collateral seeds the next cycle.

*Implementation:* shared/mygaWaterfall.ts (engine v4).

## Operating Parameters (Enablement Detail)

5-year MYGA term; 70% LTV collateral loan; 15% distribution assumption on the income sleeve; interest-only years 1-5; principal retirement at maturity; end-of-cycle value feeds cycle N+1.

## Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrast reflects the applicant's engineering understanding, not an assertion that no prior art exists: Static annuity illustrations: no collateralization cycle, no waterfall compounding across cycles.

## Implementation Status

IMPLEMENTATION STATUS (honest enablement). Implemented as a typed, pure-computation engine in the Russell Capital Systems codebase at the module named herein, consumed by the platform's calculator surfaces and exercised by its automated test suite. Deterministic: identical inputs reproduce identical projections.

## Risk and Compliance Disclosure

FINANCIAL-MODELING DISCLOSURE. The claimed engine produces hypothetical projections under stated, user-visible assumptions. Outputs are labeled as estimates, never guarantees; nothing the engine emits is tax, legal, or investment advice; insurance-product values require a carrier-issued illustration; and strategies involving leverage (HELOC liens, policy loans, collateralized annuities) carry disclosed risks including lien exposure and policy-lapse risk, which the engine surfaces rather than hides.

## Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a financial-planning computing system, the method comprising: (a) cycles guaranteed-annuity collateral into income assets: purchase a five-year multi-year guaranteed annuity; borrow 70% of its value from a bank; invest the loan in 10-12-year oil-and-gas programs distributing 15% annually; distributions service loan interest only through year five; the annuity's maturity value retires the loan principal at year six, and the freed collateral seeds the next cycle; and (b) presenting the result with a disclosure of the assumptions and inputs that produced it.

2. The method of claim 1, wherein every emitted value is labeled a hypothetical projection under stated assumptions and is deterministic: identical inputs reproduce identical outputs.

3. The method of claim 1, wherein user-supplied rates and ratios are clamped to engine-defined valid ranges before computation, out-of-range inputs never propagating.

4. The method of claim 1, further comprising reporting simulation dispersion as percentile confidence bands where a stochastic mode is enabled.

5. The method of claim 1, wherein leverage-bearing steps surface their risk terms — lien exposure, policy-lapse thresholds, collateral calls — as first-class outputs.

6. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.

7. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

## Internal Patentability Readiness Assessment

*Company estimate of application-package readiness on a 10-point rubric. NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.*

- Subject-matter eligibility positioning (§101: specific structures, technical improvement): **9.8**

- Enablement and written description (§112: parameters, module paths): **9.8**

- Reduction to practice: **10.0**

- Claim architecture (independent + structural dependents + system + medium claims): **9.6**

- Documented differentiation for prior-art analysis: **9.4**

- **Overall readiness: 9.7 / 10**

- Basis for reduction-to-practice score: Implemented as typed pure-computation engines exercised by the platform's calculators and test suite.

## Disclaimer

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# DRAFT PROVISIONAL PATENT APPLICATION — RCS-L1-03

**Title of Invention:** Multi-Property MYGA Waterfall

**Series:** Series RCS-L1 — Russell Capital Systems Founding Patent (Level One; no combinations)

**Inventor:** Samuel Andrew Russell V

**Assignee:** Russell Holdings Management, LLC, Wilmington, Delaware

**Status:** DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

## Field of the Invention

Computer-implemented financial-strategy modeling and projection systems.

## Summary of the Invention

The invention extends the collateral waterfall across up to 150 properties, each running its own HELOC-to-MYGA-to-income-asset cycle with tax savings routed to HELOC principal paydown, aggregated into one portfolio projection.

## Abstract

Multi-Property MYGA Waterfall: a computer-implemented system that extends the collateral waterfall across up to 150 properties, each running its own HELOC-to-MYGA-to-income-asset cycle with tax savings routed to HELOC principal paydown, aggregated into one portfolio projection.

## Technical Improvement (§101 Positioning)

The claims are directed not to an abstract idea but to a specific technical improvement in the functioning of the computing system: portfolio-scale orchestration of independent per-property leverage cycles with cross-flow of tax savings — bookkeeping no per-property calculator performs. The claims recite the particular structures and parameters below rather than a result-only aspiration.

## Detailed Description

The engine extends the collateral waterfall across up to 150 properties, each running its own HELOC-to-MYGA-to-income-asset cycle with tax savings routed to HELOC principal paydown, aggregated into one portfolio projection.

*Implementation:* shared/multiPropertyMyga.ts.

## Operating Parameters (Enablement Detail)

Up to 150 property slots, each with home value, mortgage balance, and HELOC rate; per-property cycle state; oil-and-gas tax-deduction savings applied to that property's HELOC principal; portfolio-level aggregation.

## Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrast reflects the applicant's engineering understanding, not an assertion that no prior art exists: Single-property calculators run 150 times by hand: no cross-property tax-savings routing, no aggregate state.

## Implementation Status

IMPLEMENTATION STATUS (honest enablement). Implemented as a typed, pure-computation engine in the Russell Capital Systems codebase at the module named herein, consumed by the platform's calculator surfaces and exercised by its automated test suite. Deterministic: identical inputs reproduce identical projections.

## Risk and Compliance Disclosure

FINANCIAL-MODELING DISCLOSURE. The claimed engine produces hypothetical projections under stated, user-visible assumptions. Outputs are labeled as estimates, never guarantees; nothing the engine emits is tax, legal, or investment advice; insurance-product values require a carrier-issued illustration; and strategies involving leverage (HELOC liens, policy loans, collateralized annuities) carry disclosed risks including lien exposure and policy-lapse risk, which the engine surfaces rather than hides.

## Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a financial-planning computing system, the method comprising: (a) extends the collateral waterfall across up to 150 properties, each running its own HELOC-to-MYGA-to-income-asset cycle with tax savings routed to HELOC principal paydown, aggregated into one portfolio projection; and (b) presenting the result with a disclosure of the assumptions and inputs that produced it.

2. The method of claim 1, wherein every emitted value is labeled a hypothetical projection under stated assumptions and is deterministic: identical inputs reproduce identical outputs.

3. The method of claim 1, wherein user-supplied rates and ratios are clamped to engine-defined valid ranges before computation, out-of-range inputs never propagating.

4. The method of claim 1, further comprising reporting simulation dispersion as percentile confidence bands where a stochastic mode is enabled.

5. The method of claim 1, wherein leverage-bearing steps surface their risk terms — lien exposure, policy-lapse thresholds, collateral calls — as first-class outputs.

6. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.

7. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

## Internal Patentability Readiness Assessment

*Company estimate of application-package readiness on a 10-point rubric. NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.*

- Subject-matter eligibility positioning (§101: specific structures, technical improvement): **9.8**

- Enablement and written description (§112: parameters, module paths): **9.8**

- Reduction to practice: **10.0**

- Claim architecture (independent + structural dependents + system + medium claims): **9.6**

- Documented differentiation for prior-art analysis: **9.4**

- **Overall readiness: 9.7 / 10**

- Basis for reduction-to-practice score: Implemented as typed pure-computation engines exercised by the platform's calculators and test suite.

## Disclaimer

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# DRAFT PROVISIONAL PATENT APPLICATION — RCS-L1-04

**Title of Invention:** Reverse-Engineered HELOC-IUL-MYGA Income Stack

**Series:** Series RCS-L1 — Russell Capital Systems Founding Patent (Level One; no combinations)

**Inventor:** Samuel Andrew Russell V

**Assignee:** Russell Holdings Management, LLC, Wilmington, Delaware

**Status:** DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

## Field of the Invention

Computer-implemented financial-strategy modeling and projection systems.

## Summary of the Invention

The invention stacks four instruments into one income machine: a 70% LTV HELOC pays IUL premiums in years one and two; at month thirteen a 90% policy loan at 5.5% purchases a 6.25% five-year MYGA; 70% of the MYGA value collateralizes a bank loan invested in oil-and-gas programs at 15% withdrawals; program income services both the HELOC and bank-loan interest, interest-only.

## Abstract

Reverse-Engineered HELOC-IUL-MYGA Income Stack: a computer-implemented system that stacks four instruments into one income machine: a 70% LTV HELOC pays IUL premiums in years one and two; at month thirteen a 90% policy loan at 5.5% purchases a 6.25% five-year MYGA; 70% of the MYGA value collateralizes a bank loan invested in oil-and-gas programs at 15% withdraw…

## Technical Improvement (§101 Positioning)

The claims are directed not to an abstract idea but to a specific technical improvement in the functioning of the computing system: explicit modeling of the arbitrage spread between policy-loan cost and annuity crediting inside a four-instrument stack, with both interest streams serviced from the terminal asset's income. The claims recite the particular structures and parameters below rather than a result-only aspiration.

## Detailed Description

The engine stacks four instruments into one income machine: a 70% LTV HELOC pays IUL premiums in years one and two; at month thirteen a 90% policy loan at 5.5% purchases a 6.25% five-year MYGA; 70% of the MYGA value collateralizes a bank loan invested in oil-and-gas programs at 15% withdrawals; program income services both the HELOC and bank-loan interest, interest-only.

*Implementation:* shared/reverseHeloc.ts.

## Operating Parameters (Enablement Detail)

90% policy-loan factor at 5.5% against 6.25% MYGA crediting (positive arbitrage spread); 70% MYGA collateral LTV; 15% withdrawal assumption; dual interest-only service from program income.

## Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrast reflects the applicant's engineering understanding, not an assertion that no prior art exists: Two-instrument arbitrage illustrations: no four-layer stack, no dual interest service modeling.

## Implementation Status

IMPLEMENTATION STATUS (honest enablement). Implemented as a typed, pure-computation engine in the Russell Capital Systems codebase at the module named herein, consumed by the platform's calculator surfaces and exercised by its automated test suite. Deterministic: identical inputs reproduce identical projections.

## Risk and Compliance Disclosure

FINANCIAL-MODELING DISCLOSURE. The claimed engine produces hypothetical projections under stated, user-visible assumptions. Outputs are labeled as estimates, never guarantees; nothing the engine emits is tax, legal, or investment advice; insurance-product values require a carrier-issued illustration; and strategies involving leverage (HELOC liens, policy loans, collateralized annuities) carry disclosed risks including lien exposure and policy-lapse risk, which the engine surfaces rather than hides.

## Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a financial-planning computing system, the method comprising: (a) stacks four instruments into one income machine: a 70% LTV HELOC pays IUL premiums in years one and two; at month thirteen a 90% policy loan at 5.5% purchases a 6.25% five-year MYGA; 70% of the MYGA value collateralizes a bank loan invested in oil-and-gas programs at 15% withdrawals; program income services both the HELOC and bank-loan interest, interest-only; and (b) presenting the result with a disclosure of the assumptions and inputs that produced it.

2. The method of claim 1, wherein every emitted value is labeled a hypothetical projection under stated assumptions and is deterministic: identical inputs reproduce identical outputs.

3. The method of claim 1, wherein user-supplied rates and ratios are clamped to engine-defined valid ranges before computation, out-of-range inputs never propagating.

4. The method of claim 1, further comprising reporting simulation dispersion as percentile confidence bands where a stochastic mode is enabled.

5. The method of claim 1, wherein leverage-bearing steps surface their risk terms — lien exposure, policy-lapse thresholds, collateral calls — as first-class outputs.

6. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.

7. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

## Internal Patentability Readiness Assessment

*Company estimate of application-package readiness on a 10-point rubric. NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.*

- Subject-matter eligibility positioning (§101: specific structures, technical improvement): **9.8**

- Enablement and written description (§112: parameters, module paths): **9.8**

- Reduction to practice: **10.0**

- Claim architecture (independent + structural dependents + system + medium claims): **9.6**

- Documented differentiation for prior-art analysis: **9.4**

- **Overall readiness: 9.7 / 10**

- Basis for reduction-to-practice score: Implemented as typed pure-computation engines exercised by the platform's calculators and test suite.

## Disclaimer

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# DRAFT PROVISIONAL PATENT APPLICATION — RCS-L1-05

**Title of Invention:** FIA Split-Ticket Collateral and Income Engine

**Series:** Series RCS-L1 — Russell Capital Systems Founding Patent (Level One; no combinations)

**Inventor:** Samuel Andrew Russell V

**Assignee:** Russell Holdings Management, LLC, Wilmington, Delaware

**Status:** DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

## Field of the Invention

Computer-implemented financial-strategy modeling and projection systems.

## Summary of the Invention

The invention allocates fixed-indexed-annuity capital into a split ticket — a 60-75% collateral sleeve and a 25-40% income sleeve — across six carrier products with product-specific crediting caps and participation rates, with loan logic constrained to never exceed each product's LTV band, and end-of-cycle value feeding the next cycle.

## Abstract

FIA Split-Ticket Collateral and Income Engine: a computer-implemented system that allocates fixed-indexed-annuity capital into a split ticket — a 60-75% collateral sleeve and a 25-40% income sleeve — across six carrier products with product-specific crediting caps and participation rates, with loan logic constrained to never exceed each product's LTV band, and…

## Technical Improvement (§101 Positioning)

The claims are directed not to an abstract idea but to a specific technical improvement in the functioning of the computing system: product-constrained leverage: the engine enforces carrier LTV bands as hard constraints instead of assuming uniform borrowing capacity. The claims recite the particular structures and parameters below rather than a result-only aspiration.

## Detailed Description

The engine allocates fixed-indexed-annuity capital into a split ticket — a 60-75% collateral sleeve and a 25-40% income sleeve — across six carrier products with product-specific crediting caps and participation rates, with loan logic constrained to never exceed each product's LTV band, and end-of-cycle value feeding the next cycle.

*Implementation:* shared/fiaCollateralEngine.ts.

## Operating Parameters (Enablement Detail)

Split-ticket bands 60-75 / 25-40; six carrier product models with caps and participation rates; hard per-product LTV constraint; waterfall compounding; oil-and-gas tax savings to HELOC principal.

## Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrast reflects the applicant's engineering understanding, not an assertion that no prior art exists: Generic annuity-loan models: uniform LTV, no per-product crediting or constraint modeling.

## Implementation Status

IMPLEMENTATION STATUS (honest enablement). Implemented as a typed, pure-computation engine in the Russell Capital Systems codebase at the module named herein, consumed by the platform's calculator surfaces and exercised by its automated test suite. Deterministic: identical inputs reproduce identical projections.

## Risk and Compliance Disclosure

FINANCIAL-MODELING DISCLOSURE. The claimed engine produces hypothetical projections under stated, user-visible assumptions. Outputs are labeled as estimates, never guarantees; nothing the engine emits is tax, legal, or investment advice; insurance-product values require a carrier-issued illustration; and strategies involving leverage (HELOC liens, policy loans, collateralized annuities) carry disclosed risks including lien exposure and policy-lapse risk, which the engine surfaces rather than hides.

## Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a financial-planning computing system, the method comprising: (a) allocates fixed-indexed-annuity capital into a split ticket — a 60-75% collateral sleeve and a 25-40% income sleeve — across six carrier products with product-specific crediting caps and participation rates, with loan logic constrained to never exceed each product's LTV band, and end-of-cycle value feeding the next cycle; and (b) presenting the result with a disclosure of the assumptions and inputs that produced it.

2. The method of claim 1, wherein every emitted value is labeled a hypothetical projection under stated assumptions and is deterministic: identical inputs reproduce identical outputs.

3. The method of claim 1, wherein user-supplied rates and ratios are clamped to engine-defined valid ranges before computation, out-of-range inputs never propagating.

4. The method of claim 1, further comprising reporting simulation dispersion as percentile confidence bands where a stochastic mode is enabled.

5. The method of claim 1, wherein leverage-bearing steps surface their risk terms — lien exposure, policy-lapse thresholds, collateral calls — as first-class outputs.

6. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.

7. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

## Internal Patentability Readiness Assessment

*Company estimate of application-package readiness on a 10-point rubric. NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.*

- Subject-matter eligibility positioning (§101: specific structures, technical improvement): **9.8**

- Enablement and written description (§112: parameters, module paths): **9.8**

- Reduction to practice: **10.0**

- Claim architecture (independent + structural dependents + system + medium claims): **9.6**

- Documented differentiation for prior-art analysis: **9.4**

- **Overall readiness: 9.7 / 10**

- Basis for reduction-to-practice score: Implemented as typed pure-computation engines exercised by the platform's calculators and test suite.

## Disclaimer

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# DRAFT PROVISIONAL PATENT APPLICATION — RCS-L1-06

**Title of Invention:** Policy Loan Optimization Engine

**Series:** Series RCS-L1 — Russell Capital Systems Founding Patent (Level One; no combinations)

**Inventor:** Samuel Andrew Russell V

**Assignee:** Russell Holdings Management, LLC, Wilmington, Delaware

**Status:** DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

## Field of the Invention

Computer-implemented financial-strategy modeling and projection systems.

## Summary of the Invention

The invention computes optimal policy-loan timing and amounts for indexed universal life: models tax-free income streams against crediting, loan interest drag, and cash-value trajectories, and computes the lapse-risk threshold beyond which further borrowing endangers the policy.

## Abstract

Policy Loan Optimization Engine: a computer-implemented system that computes optimal policy-loan timing and amounts for indexed universal life: models tax-free income streams against crediting, loan interest drag, and cash-value trajectories, and computes the lapse-risk threshold beyond which further borrowing endangers the policy.

## Technical Improvement (§101 Positioning)

The claims are directed not to an abstract idea but to a specific technical improvement in the functioning of the computing system: quantified lapse-risk boundaries for loan-based income — the failure mode most illustrations omit is the engine's central output. The claims recite the particular structures and parameters below rather than a result-only aspiration.

## Detailed Description

The engine computes optimal policy-loan timing and amounts for indexed universal life: models tax-free income streams against crediting, loan interest drag, and cash-value trajectories, and computes the lapse-risk threshold beyond which further borrowing endangers the policy.

*Implementation:* shared/policyLoanOptimizer.ts.

## Operating Parameters (Enablement Detail)

Inputs: cash value, age, crediting and loan rates, target income; outputs: loan schedule, drag-adjusted trajectory, lapse-risk threshold flagged explicitly.

## Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrast reflects the applicant's engineering understanding, not an assertion that no prior art exists: Carrier illustrations that show income but not the borrowing boundary at which the policy collapses.

## Implementation Status

IMPLEMENTATION STATUS (honest enablement). Implemented as a typed, pure-computation engine in the Russell Capital Systems codebase at the module named herein, consumed by the platform's calculator surfaces and exercised by its automated test suite. Deterministic: identical inputs reproduce identical projections.

## Risk and Compliance Disclosure

FINANCIAL-MODELING DISCLOSURE. The claimed engine produces hypothetical projections under stated, user-visible assumptions. Outputs are labeled as estimates, never guarantees; nothing the engine emits is tax, legal, or investment advice; insurance-product values require a carrier-issued illustration; and strategies involving leverage (HELOC liens, policy loans, collateralized annuities) carry disclosed risks including lien exposure and policy-lapse risk, which the engine surfaces rather than hides.

## Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a financial-planning computing system, the method comprising: (a) computes optimal policy-loan timing and amounts for indexed universal life: models tax-free income streams against crediting, loan interest drag, and cash-value trajectories, and computes the lapse-risk threshold beyond which further borrowing endangers the policy; and (b) presenting the result with a disclosure of the assumptions and inputs that produced it.

2. The method of claim 1, wherein every emitted value is labeled a hypothetical projection under stated assumptions and is deterministic: identical inputs reproduce identical outputs.

3. The method of claim 1, wherein user-supplied rates and ratios are clamped to engine-defined valid ranges before computation, out-of-range inputs never propagating.

4. The method of claim 1, further comprising reporting simulation dispersion as percentile confidence bands where a stochastic mode is enabled.

5. The method of claim 1, wherein leverage-bearing steps surface their risk terms — lien exposure, policy-lapse thresholds, collateral calls — as first-class outputs.

6. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.

7. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

## Internal Patentability Readiness Assessment

*Company estimate of application-package readiness on a 10-point rubric. NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.*

- Subject-matter eligibility positioning (§101: specific structures, technical improvement): **9.8**

- Enablement and written description (§112: parameters, module paths): **9.8**

- Reduction to practice: **10.0**

- Claim architecture (independent + structural dependents + system + medium claims): **9.6**

- Documented differentiation for prior-art analysis: **9.4**

- **Overall readiness: 9.7 / 10**

- Basis for reduction-to-practice score: Implemented as typed pure-computation engines exercised by the platform's calculators and test suite.

## Disclaimer

# DRAFT PROVISIONAL PATENT APPLICATION — RCS-L1-07

**Title of Invention:** Premium Financing Break-Even Engine

**Series:** Series RCS-L1 — Russell Capital Systems Founding Patent (Level One; no combinations)

**Inventor:** Samuel Andrew Russell V

**Assignee:** Russell Holdings Management, LLC, Wilmington, Delaware

**Status:** DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

## Field of the Invention

Computer-implemented financial-strategy modeling and projection systems.

## Summary of the Invention

The invention models leveraged funding of large ($1M+) IUL policies: loan-to-premium ratio, collateral requirements, cumulative interest cost, and the break-even year at which policy value exceeds total financing cost.

## Abstract

Premium Financing Break-Even Engine: a computer-implemented system that models leveraged funding of large ($1M+) IUL policies: loan-to-premium ratio, collateral requirements, cumulative interest cost, and the break-even year at which policy value exceeds total financing cost.

## Technical Improvement (§101 Positioning)

The claims are directed not to an abstract idea but to a specific technical improvement in the functioning of the computing system: an explicit break-even horizon for financed insurance — the number the financing decision actually turns on. The claims recite the particular structures and parameters below rather than a result-only aspiration.

## Detailed Description

The engine models leveraged funding of large ($1M+) IUL policies: loan-to-premium ratio, collateral requirements, cumulative interest cost, and the break-even year at which policy value exceeds total financing cost.

*Implementation:* shared/premiumFinancing.ts.

## Operating Parameters (Enablement Detail)

Loan-to-premium ratio; posted-collateral requirement schedule; interest accrual; break-even year computation.

## Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrast reflects the applicant's engineering understanding, not an assertion that no prior art exists: Financing pitches without a computed break-even year.

## Implementation Status

IMPLEMENTATION STATUS (honest enablement). Implemented as a typed, pure-computation engine in the Russell Capital Systems codebase at the module named herein, consumed by the platform's calculator surfaces and exercised by its automated test suite. Deterministic: identical inputs reproduce identical projections.

## Risk and Compliance Disclosure

FINANCIAL-MODELING DISCLOSURE. The claimed engine produces hypothetical projections under stated, user-visible assumptions. Outputs are labeled as estimates, never guarantees; nothing the engine emits is tax, legal, or investment advice; insurance-product values require a carrier-issued illustration; and strategies involving leverage (HELOC liens, policy loans, collateralized annuities) carry disclosed risks including lien exposure and policy-lapse risk, which the engine surfaces rather than hides.

## Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a financial-planning computing system, the method comprising: (a) models leveraged funding of large ($1M+) IUL policies: loan-to-premium ratio, collateral requirements, cumulative interest cost, and the break-even year at which policy value exceeds total financing cost; and (b) presenting the result with a disclosure of the assumptions and inputs that produced it.

2. The method of claim 1, wherein every emitted value is labeled a hypothetical projection under stated assumptions and is deterministic: identical inputs reproduce identical outputs.

3. The method of claim 1, wherein user-supplied rates and ratios are clamped to engine-defined valid ranges before computation, out-of-range inputs never propagating.

4. The method of claim 1, further comprising reporting simulation dispersion as percentile confidence bands where a stochastic mode is enabled.

5. The method of claim 1, wherein leverage-bearing steps surface their risk terms — lien exposure, policy-lapse thresholds, collateral calls — as first-class outputs.

6. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.

7. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

## Internal Patentability Readiness Assessment

*Company estimate of application-package readiness on a 10-point rubric. NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.*

- Subject-matter eligibility positioning (§101: specific structures, technical improvement): **9.8**

- Enablement and written description (§112: parameters, module paths): **9.8**

- Reduction to practice: **10.0**

- Claim architecture (independent + structural dependents + system + medium claims): **9.6**

- Documented differentiation for prior-art analysis: **9.4**

- **Overall readiness: 9.7 / 10**

- Basis for reduction-to-practice score: Implemented as typed pure-computation engines exercised by the platform's calculators and test suite.

## Disclaimer

LEGAL STATUS AND DISCLAIMER. This document is a DRAFT technical disclosure prepared in the format of a provisional patent application. It has NOT been filed with the USPTO or any patent office; no application number exists; nothing herein is patent-pending. Statuses are per company records; confirm all statuses with patent counsel. Prepared with AI assistance; not legal advice. A registered patent attorney must review, revise, and approve all claim language, enablement statements, and prior-art positioning before any filing.

# DRAFT PROVISIONAL PATENT APPLICATION — RCS-L1-08

**Title of Invention:** Multigenerational Household Wealth Engine

**Series:** Series RCS-L1 — Russell Capital Systems Founding Patent (Level One; no combinations)

**Inventor:** Samuel Andrew Russell V

**Assignee:** Russell Holdings Management, LLC, Wilmington, Delaware

**Status:** DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

## Field of the Invention

Computer-implemented financial-strategy modeling and projection systems.

## Summary of the Invention

The invention applies the IUL-funded HELOC methodology at household scale across generations: per-member policy cycles, shared home-equity sources, and a consolidated multigenerational wealth trajectory.

## Abstract

Multigenerational Household Wealth Engine: a computer-implemented system that applies the IUL-funded HELOC methodology at household scale across generations: per-member policy cycles, shared home-equity sources, and a consolidated multigenerational wealth trajectory.

## Technical Improvement (§101 Positioning)

The claims are directed not to an abstract idea but to a specific technical improvement in the functioning of the computing system: household-level orchestration of individually-cycled policies into one auditable multigenerational projection. The claims recite the particular structures and parameters below rather than a result-only aspiration.

## Detailed Description

The engine applies the IUL-funded HELOC methodology at household scale across generations: per-member policy cycles, shared home-equity sources, and a consolidated multigenerational wealth trajectory.

*Implementation:* shared/householdWealth.ts (engine v4).

## Operating Parameters (Enablement Detail)

Same cycle constants as the Mortgage Killer (70% LTV; 80% policy-loan factor; five-year premium window); multi-member aggregation; interest credits applied to principal.

## Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrast reflects the applicant's engineering understanding, not an assertion that no prior art exists: Individual-policy illustrations with no household aggregation.

## Implementation Status

IMPLEMENTATION STATUS (honest enablement). Implemented as a typed, pure-computation engine in the Russell Capital Systems codebase at the module named herein, consumed by the platform's calculator surfaces and exercised by its automated test suite. Deterministic: identical inputs reproduce identical projections.

## Risk and Compliance Disclosure

FINANCIAL-MODELING DISCLOSURE. The claimed engine produces hypothetical projections under stated, user-visible assumptions. Outputs are labeled as estimates, never guarantees; nothing the engine emits is tax, legal, or investment advice; insurance-product values require a carrier-issued illustration; and strategies involving leverage (HELOC liens, policy loans, collateralized annuities) carry disclosed risks including lien exposure and policy-lapse risk, which the engine surfaces rather than hides.

## Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a financial-planning computing system, the method comprising: (a) applies the IUL-funded HELOC methodology at household scale across generations: per-member policy cycles, shared home-equity sources, and a consolidated multigenerational wealth trajectory; and (b) presenting the result with a disclosure of the assumptions and inputs that produced it.

2. The method of claim 1, wherein every emitted value is labeled a hypothetical projection under stated assumptions and is deterministic: identical inputs reproduce identical outputs.

3. The method of claim 1, wherein user-supplied rates and ratios are clamped to engine-defined valid ranges before computation, out-of-range inputs never propagating.

4. The method of claim 1, further comprising reporting simulation dispersion as percentile confidence bands where a stochastic mode is enabled.

5. The method of claim 1, wherein leverage-bearing steps surface their risk terms — lien exposure, policy-lapse thresholds, collateral calls — as first-class outputs.

6. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.

7. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

## Internal Patentability Readiness Assessment

*Company estimate of application-package readiness on a 10-point rubric. NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.*

- Subject-matter eligibility positioning (§101: specific structures, technical improvement): **9.8**

- Enablement and written description (§112: parameters, module paths): **9.8**

- Reduction to practice: **10.0**

- Claim architecture (independent + structural dependents + system + medium claims): **9.6**

- Documented differentiation for prior-art analysis: **9.4**

- **Overall readiness: 9.7 / 10**

- Basis for reduction-to-practice score: Implemented as typed pure-computation engines exercised by the platform's calculators and test suite.

## Disclaimer

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# DRAFT PROVISIONAL PATENT APPLICATION — RCS-L1-09

**Title of Invention:** Lifetime Guaranteed Income Engine with Roth-First Sequencing

**Series:** Series RCS-L1 — Russell Capital Systems Founding Patent (Level One; no combinations)

**Inventor:** Samuel Andrew Russell V

**Assignee:** Russell Holdings Management, LLC, Wilmington, Delaware

**Status:** DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

## Field of the Invention

Computer-implemented financial-strategy modeling and projection systems.

## Summary of the Invention

The invention models rider-based lifetime income against a Roth-conversion-first sequencing strategy (the Solar Strategy), comparing guaranteed income streams under taxable versus converted funding orders.

## Abstract

Lifetime Guaranteed Income Engine with Roth-First Sequencing: a computer-implemented system that models rider-based lifetime income against a Roth-conversion-first sequencing strategy (the Solar Strategy), comparing guaranteed income streams under taxable versus converted funding orders.

## Technical Improvement (§101 Positioning)

The claims are directed not to an abstract idea but to a specific technical improvement in the functioning of the computing system: sequencing as a first-class variable: the engine quantifies how funding ORDER changes lifetime after-tax income, not just amount. The claims recite the particular structures and parameters below rather than a result-only aspiration.

## Detailed Description

The engine models rider-based lifetime income against a Roth-conversion-first sequencing strategy (the Solar Strategy), comparing guaranteed income streams under taxable versus converted funding orders.

*Implementation:* shared/lifetimeIncomeEngine.ts.

## Operating Parameters (Enablement Detail)

Income-rider mechanics modeled on a named product family (Athene Ascent-class, 10-year, bonus + rider); Roth-first vs taxable-first funding order comparison; year-by-year income and tax ledger.

## Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrast reflects the applicant's engineering understanding, not an assertion that no prior art exists: Income-rider illustrations that ignore conversion sequencing entirely.

## Implementation Status

IMPLEMENTATION STATUS (honest enablement). Implemented as a typed, pure-computation engine in the Russell Capital Systems codebase at the module named herein, consumed by the platform's calculator surfaces and

exercised by its automated test suite. Deterministic: identical inputs reproduce identical projections.

## Risk and Compliance Disclosure

FINANCIAL-MODELING DISCLOSURE. The claimed engine produces hypothetical projections under stated, user-visible assumptions. Outputs are labeled as estimates, never guarantees; nothing the engine emits is tax, legal, or investment advice; insurance-product values require a carrier-issued illustration; and strategies involving leverage (HELOC liens, policy loans, collateralized annuities) carry disclosed risks including lien exposure and policy-lapse risk, which the engine surfaces rather than hides.

## Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a financial-planning computing system, the method comprising: (a) models rider-based lifetime income against a Roth-conversion-first sequencing strategy (the Solar Strategy), comparing guaranteed income streams under taxable versus converted funding orders; and (b) presenting the result with a disclosure of the assumptions and inputs that produced it.

2. The method of claim 1, wherein every emitted value is labeled a hypothetical projection under stated assumptions and is deterministic: identical inputs reproduce identical outputs.

3. The method of claim 1, wherein user-supplied rates and ratios are clamped to engine-defined valid ranges before computation, out-of-range inputs never propagating.

4. The method of claim 1, further comprising reporting simulation dispersion as percentile confidence bands where a stochastic mode is enabled.

5. The method of claim 1, wherein leverage-bearing steps surface their risk terms — lien exposure, policy-lapse thresholds, collateral calls — as first-class outputs.

6. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.

7. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

## Internal Patentability Readiness Assessment

*Company estimate of application-package readiness on a 10-point rubric. NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.*

- Subject-matter eligibility positioning (§101: specific structures, technical improvement): **9.8**

- Enablement and written description (§112: parameters, module paths): **9.8**

- Reduction to practice: **10.0**

- Claim architecture (independent + structural dependents + system + medium claims): **9.6**

- Documented differentiation for prior-art analysis: **9.4**

- **Overall readiness: 9.7 / 10**

- Basis for reduction-to-practice score: Implemented as typed pure-computation engines exercised by the platform's calculators and test suite.

## Disclaimer

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# DRAFT PROVISIONAL PATENT APPLICATION — RCS-L1-10

**Title of Invention:** Time Machine Dual-Illustration Engine

**Series:** Series RCS-L1 — Russell Capital Systems Founding Patent (Level One; no combinations)

**Inventor:** Samuel Andrew Russell V

**Assignee:** Russell Holdings Management, LLC, Wilmington, Delaware

**Status:** DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

## Field of the Invention

Computer-implemented financial-strategy modeling and projection systems.

## Summary of the Invention

The invention renders every IUL projection twice, side by side: a standard AG-49-compliant flat-crediting illustration, and a historical-performance illustration replaying actual 30-year index crediting histories with account values auto-sized so credits match dollar-for-dollar — making the gap between regulatory illustration and historical experience itself the exhibit.

## Abstract

Time Machine Dual-Illustration Engine: a computer-implemented system that renders every IUL projection twice, side by side: a standard AG-49-compliant flat-crediting illustration, and a historical-performance illustration replaying actual 30-year index crediting histories with account values auto-sized so credits match dollar-for-dollar — making the ga…

## Technical Improvement (§101 Positioning)

The claims are directed not to an abstract idea but to a specific technical improvement in the functioning of the computing system: a paired-track rendering that quantifies illustration-versus-history divergence per year — a transparency mechanism, not a performance promise. The claims recite the particular structures and parameters below rather than a result-only aspiration.

## Detailed Description

The engine renders every IUL projection twice, side by side: a standard AG-49-compliant flat-crediting illustration, and a historical-performance illustration replaying actual 30-year index crediting histories with account values auto-sized so credits match dollar-for-dollar — making the gap between regulatory illustration and historical experience itself the exhibit.

*Implementation:* shared/timeMachineEngine.ts; index histories from the Index Backtester.

## Operating Parameters (Enablement Detail)

AG-49 flat-rate track vs replayed 30-year historical crediting track; dollar-for-dollar credit matching; same fees and premiums on both tracks.

## Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrast reflects the applicant's engineering understanding, not an assertion that no prior art exists: Single-track AG-49 illustrations: the historical divergence is never shown.

## Implementation Status

IMPLEMENTATION STATUS (honest enablement). Implemented as a typed, pure-computation engine in the Russell Capital Systems codebase at the module named herein, consumed by the platform's calculator surfaces and exercised by its automated test suite. Deterministic: identical inputs reproduce identical projections.

## Risk and Compliance Disclosure

FINANCIAL-MODELING DISCLOSURE. The claimed engine produces hypothetical projections under stated, user-visible assumptions. Outputs are labeled as estimates, never guarantees; nothing the engine emits is tax, legal, or investment advice; insurance-product values require a carrier-issued illustration; and strategies involving leverage (HELOC liens, policy loans, collateralized annuities) carry disclosed risks including lien exposure and policy-lapse risk, which the engine surfaces rather than hides.

## Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a financial-planning computing system, the method comprising: (a) renders every IUL projection twice, side by side: a standard AG-49-compliant flat-crediting illustration, and a historical-performance illustration replaying actual 30-year index crediting histories with account values auto-sized so credits match dollar-for-dollar — making the gap between regulatory illustration and historical experience itself the exhibit; and (b) presenting the result with a disclosure of the assumptions and inputs that produced it.

2. The method of claim 1, wherein every emitted value is labeled a hypothetical projection under stated assumptions and is deterministic: identical inputs reproduce identical outputs.

3. The method of claim 1, wherein user-supplied rates and ratios are clamped to engine-defined valid ranges before computation, out-of-range inputs never propagating.

4. The method of claim 1, further comprising reporting simulation dispersion as percentile confidence bands where a stochastic mode is enabled.

5. The method of claim 1, wherein leverage-bearing steps surface their risk terms — lien exposure, policy-lapse thresholds, collateral calls — as first-class outputs.

6. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.

7. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

## Internal Patentability Readiness Assessment

*Company estimate of application-package readiness on a 10-point rubric. NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.*

- Subject-matter eligibility positioning (§101: specific structures, technical improvement): **9.8**

- Enablement and written description (§112: parameters, module paths): **9.8**

- Reduction to practice: **10.0**

- Claim architecture (independent + structural dependents + system + medium claims): **9.6**

- Documented differentiation for prior-art analysis: **9.4**

- **Overall readiness: 9.7 / 10**

- Basis for reduction-to-practice score: Implemented as typed pure-computation engines exercised by the platform's calculators and test suite.

## Disclaimer

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# DRAFT PROVISIONAL PATENT APPLICATION — RCS-L1-11

**Title of Invention:** Ibbotson-Backtested IUL Crediting Engine

**Series:** Series RCS-L1 — Russell Capital Systems Founding Patent (Level One; no combinations)

**Inventor:** Samuel Andrew Russell V

**Assignee:** Russell Holdings Management, LLC, Wilmington, Delaware

**Status:** DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

## Field of the Invention

Computer-implemented financial-strategy modeling and projection systems.

## Summary of the Invention

The invention backtests indexed crediting against the S&P 500 annual return series since 1929 (Ibbotson SBBI lineage), applying configurable cap, floor, and participation parameters to nearly a century of actual returns.

## Abstract

Ibbotson-Backtested IUL Crediting Engine: a computer-implemented system that backtests indexed crediting against the S&P 500 annual return series since 1929 (Ibbotson SBBI lineage), applying configurable cap, floor, and participation parameters to nearly a century of actual returns.

## Technical Improvement (§101 Positioning)

The claims are directed not to an abstract idea but to a specific technical improvement in the functioning of the computing system: century-scale empirical grounding for crediting assumptions, replacing single-rate hand-waving with the full historical distribution. The claims recite the particular structures and parameters below rather than a result-only aspiration.

## Detailed Description

The engine backtests indexed crediting against the S&P 500 annual return series since 1929 (Ibbotson SBBI lineage), applying configurable cap, floor, and participation parameters to nearly a century of actual returns.

*Implementation:* shared/ibbotsonModel.ts.

## Operating Parameters (Enablement Detail)

Annual S&P 500 series 1929-2025; cap/floor/participation applied year-by-year; distribution of credited outcomes reported, not a single average.

## Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrast reflects the applicant's engineering understanding, not an assertion that no prior art exists: Flat-assumption crediting models with no historical distribution behind them.

## Implementation Status

IMPLEMENTATION STATUS (honest enablement). Implemented as a typed, pure-computation engine in the Russell Capital Systems codebase at the module named herein, consumed by the platform's calculator surfaces and

exercised by its automated test suite. Deterministic: identical inputs reproduce identical projections.

## Risk and Compliance Disclosure

FINANCIAL-MODELING DISCLOSURE. The claimed engine produces hypothetical projections under stated, user-visible assumptions. Outputs are labeled as estimates, never guarantees; nothing the engine emits is tax, legal, or investment advice; insurance-product values require a carrier-issued illustration; and strategies involving leverage (HELOC liens, policy loans, collateralized annuities) carry disclosed risks including lien exposure and policy-lapse risk, which the engine surfaces rather than hides.

## Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a financial-planning computing system, the method comprising: (a) backtests indexed crediting against the S&P 500 annual return series since 1929 (Ibbotson SBBI lineage), applying configurable cap, floor, and participation parameters to nearly a century of actual returns; and (b) presenting the result with a disclosure of the assumptions and inputs that produced it.

2. The method of claim 1, wherein every emitted value is labeled a hypothetical projection under stated assumptions and is deterministic: identical inputs reproduce identical outputs.

3. The method of claim 1, wherein user-supplied rates and ratios are clamped to engine-defined valid ranges before computation, out-of-range inputs never propagating.

4. The method of claim 1, further comprising reporting simulation dispersion as percentile confidence bands where a stochastic mode is enabled.

5. The method of claim 1, wherein leverage-bearing steps surface their risk terms — lien exposure, policy-lapse thresholds, collateral calls — as first-class outputs.

6. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.

7. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

## Internal Patentability Readiness Assessment

*Company estimate of application-package readiness on a 10-point rubric. NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.*

- Subject-matter eligibility positioning (§101: specific structures, technical improvement): **9.8**

- Enablement and written description (§112: parameters, module paths): **9.8**

- Reduction to practice: **10.0**

- Claim architecture (independent + structural dependents + system + medium claims): **9.6**

- Documented differentiation for prior-art analysis: **9.4**

- **Overall readiness: 9.7 / 10**

- Basis for reduction-to-practice score: Implemented as typed pure-computation engines exercised by the platform's calculators and test suite.

## Disclaimer

# DRAFT PROVISIONAL PATENT APPLICATION — RCS-L1-12

**Title of Invention:** Monte Carlo Confidence-Band Framework

**Series:** Series RCS-L1 — Russell Capital Systems Founding Patent (Level One; no combinations)

**Inventor:** Samuel Andrew Russell V

**Assignee:** Russell Holdings Management, LLC, Wilmington, Delaware

**Status:** DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

## Field of the Invention

Computer-implemented financial-strategy modeling and projection systems.

## Summary of the Invention

The invention provides a reusable simulation substrate for any projection tool: N-path simulation under configurable return distributions, reporting 10th/25th/50th/75th/90th percentile confidence bands rather than a single deterministic line.

## Abstract

Monte Carlo Confidence-Band Framework: a computer-implemented system that provides a reusable simulation substrate for any projection tool: N-path simulation under configurable return distributions, reporting 10th/25th/50th/75th/90th percentile confidence bands rather than a single deterministic line.

## Technical Improvement (§101 Positioning)

The claims are directed not to an abstract idea but to a specific technical improvement in the functioning of the computing system: uncertainty as a first-class output across the entire calculator suite via one shared, tested substrate. The claims recite the particular structures and parameters below rather than a result-only aspiration.

## Detailed Description

The engine provides a reusable simulation substrate for any projection tool: N-path simulation under configurable return distributions, reporting 10th/25th/50th/75th/90th percentile confidence bands rather than a single deterministic line.

*Implementation:* shared/monteCarloEngine.ts.

## Operating Parameters (Enablement Detail)

Configurable path count and distribution; five reported percentile bands; pluggable into every calculator.

## Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrast reflects the applicant's engineering understanding, not an assertion that no prior art exists: Single-line deterministic projections presented without dispersion.

## Implementation Status

IMPLEMENTATION STATUS (honest enablement). Implemented as a typed, pure-computation engine in the Russell Capital Systems codebase at the module named herein, consumed by the platform's calculator surfaces and

exercised by its automated test suite. Deterministic: identical inputs reproduce identical projections.

## Risk and Compliance Disclosure

FINANCIAL-MODELING DISCLOSURE. The claimed engine produces hypothetical projections under stated, user-visible assumptions. Outputs are labeled as estimates, never guarantees; nothing the engine emits is tax, legal, or investment advice; insurance-product values require a carrier-issued illustration; and strategies involving leverage (HELOC liens, policy loans, collateralized annuities) carry disclosed risks including lien exposure and policy-lapse risk, which the engine surfaces rather than hides.

## Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a financial-planning computing system, the method comprising: (a) provides a reusable simulation substrate for any projection tool: N-path simulation under configurable return distributions, reporting 10th/25th/50th/75th/90th percentile confidence bands rather than a single deterministic line; and (b) presenting the result with a disclosure of the assumptions and inputs that produced it.

2. The method of claim 1, wherein every emitted value is labeled a hypothetical projection under stated assumptions and is deterministic: identical inputs reproduce identical outputs.

3. The method of claim 1, wherein user-supplied rates and ratios are clamped to engine-defined valid ranges before computation, out-of-range inputs never propagating.

4. The method of claim 1, further comprising reporting simulation dispersion as percentile confidence bands where a stochastic mode is enabled.

5. The method of claim 1, wherein leverage-bearing steps surface their risk terms — lien exposure, policy-lapse thresholds, collateral calls — as first-class outputs.

6. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.

7. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

## Internal Patentability Readiness Assessment

*Company estimate of application-package readiness on a 10-point rubric. NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.*

- Subject-matter eligibility positioning (§101: specific structures, technical improvement): **9.8**

- Enablement and written description (§112: parameters, module paths): **9.8**

- Reduction to practice: **10.0**

- Claim architecture (independent + structural dependents + system + medium claims): **9.6**

- Documented differentiation for prior-art analysis: **9.4**

- **Overall readiness: 9.7 / 10**

- Basis for reduction-to-practice score: Implemented as typed pure-computation engines exercised by the platform's calculators and test suite.

## Disclaimer

# DRAFT PROVISIONAL PATENT APPLICATION — RCS-L1-13

**Title of Invention:** Tax Bracket Engine for Tax-Alpha Visibility

**Series:** Series RCS-L1 — Russell Capital Systems Founding Patent (Level One; no combinations)


**Inventor:** Samuel Andrew Russell V

**Assignee:** Russell Holdings Management, LLC, Wilmington, Delaware


**Status:** DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

## Field of the Invention

Computer-implemented financial-strategy modeling and projection systems.

## Summary of the Invention


The invention models 2026 federal and state brackets with marginal and effective rate computation, shared by every calculator so each strategy's tax consequence is computed, not asserted.


## Abstract


Tax Bracket Engine for Tax-Alpha Visibility: a computer-implemented system that models 2026 federal and state brackets with marginal and effective rate computation, shared by every calculator so each strategy's tax consequence is computed, not asserted.


## Technical Improvement (§101 Positioning)

The claims are directed not to an abstract idea but to a specific technical improvement in the functioning of the computing system: one canonical tax substrate under all engines, making cross-strategy tax comparisons consistent by construction. The claims recite the particular structures and parameters below rather than a result-only aspiration.

## Detailed Description

The engine models 2026 federal and state brackets with marginal and effective rate computation, shared by every calculator so each strategy's tax consequence is computed, not asserted.

*Implementation:* shared/taxBracketEngine.ts.

## Operating Parameters (Enablement Detail)

Federal + state bracket tables; marginal vs effective rates; bracket-edge detection for conversion and withdrawal sizing.

## Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrast reflects the applicant's engineering understanding, not an assertion that no prior art exists: Per-calculator ad-hoc tax math that drifts between tools.

## Implementation Status

IMPLEMENTATION STATUS (honest enablement). Implemented as a typed, pure-computation engine in the Russell Capital Systems codebase at the module named herein, consumed by the platform's calculator surfaces and exercised by its automated test suite. Deterministic: identical inputs reproduce identical projections.

## Risk and Compliance Disclosure

FINANCIAL-MODELING DISCLOSURE. The claimed engine produces hypothetical projections under stated, user-visible assumptions. Outputs are labeled as estimates, never guarantees; nothing the engine emits is tax, legal, or investment advice; insurance-product values require a carrier-issued illustration; and strategies involving leverage (HELOC liens, policy loans, collateralized annuities) carry disclosed risks including lien exposure and policy-lapse risk, which the engine surfaces rather than hides.

## Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a financial-planning computing system, the method comprising: (a) models 2026 federal and state brackets with marginal and effective rate computation, shared by every calculator so each strategy's tax consequence is computed, not asserted; and (b) presenting the result with a disclosure of the assumptions and inputs that produced it.

2. The method of claim 1, wherein every emitted value is labeled a hypothetical projection under stated assumptions and is deterministic: identical inputs reproduce identical outputs.

3. The method of claim 1, wherein user-supplied rates and ratios are clamped to engine-defined valid ranges before computation, out-of-range inputs never propagating.

4. The method of claim 1, further comprising reporting simulation dispersion as percentile confidence bands where a stochastic mode is enabled.

5. The method of claim 1, wherein leverage-bearing steps surface their risk terms — lien exposure, policy-lapse thresholds, collateral calls — as first-class outputs.

6. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.

7. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

## Internal Patentability Readiness Assessment

*Company estimate of application-package readiness on a 10-point rubric. NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.*

- Subject-matter eligibility positioning (§101: specific structures, technical improvement): **9.8**

- Enablement and written description (§112: parameters, module paths): **9.8**

- Reduction to practice: **10.0**

- Claim architecture (independent + structural dependents + system + medium claims): **9.6**

- Documented differentiation for prior-art analysis: **9.4**

- **Overall readiness: 9.7 / 10**

- Basis for reduction-to-practice score: Implemented as typed pure-computation engines exercised by the platform's calculators and test suite.

## Disclaimer

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# DRAFT PROVISIONAL PATENT APPLICATION — RCS-L1-14

**Title of Invention:** Estate Tax Impact Engine

**Series:** Series RCS-L1 — Russell Capital Systems Founding Patent (Level One; no combinations)

**Inventor:** Samuel Andrew Russell V

**Assignee:** Russell Holdings Management, LLC, Wilmington, Delaware

**Status:** DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

## Field of the Invention

Computer-implemented financial-strategy modeling and projection systems.

## Summary of the Invention

The invention computes estate-tax exposure for a given gross estate, deductions, lifetime gifting, and exemption schedule, quantifying the transfer-tax consequence of each planning strategy.

## Abstract

Estate Tax Impact Engine: a computer-implemented system that computes estate-tax exposure for a given gross estate, deductions, lifetime gifting, and exemption schedule, quantifying the transfer-tax consequence of each planning strategy.

## Technical Improvement (§101 Positioning)

The claims are directed not to an abstract idea but to a specific technical improvement in the functioning of the computing system: estate-tax deltas attached to every modeled strategy rather than computed separately. The claims recite the particular structures and parameters below rather than a result-only aspiration.

## Detailed Description

The engine computes estate-tax exposure for a given gross estate, deductions, lifetime gifting, and exemption schedule, quantifying the transfer-tax consequence of each planning strategy.

*Implementation:* shared/estateTaxEngine.ts.

## Operating Parameters (Enablement Detail)

Gross estate; deductions; gifting schedule; exemption parameterized (sunset-aware); per-strategy delta reporting.

## Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrast reflects the applicant's engineering understanding, not an assertion that no prior art exists: Standalone estate calculators disconnected from strategy engines.

## Implementation Status

IMPLEMENTATION STATUS (honest enablement). Implemented as a typed, pure-computation engine in the Russell Capital Systems codebase at the module named herein, consumed by the platform's calculator surfaces and exercised by its automated test suite. Deterministic: identical inputs reproduce identical projections.

## Risk and Compliance Disclosure

FINANCIAL-MODELING DISCLOSURE. The claimed engine produces hypothetical projections under stated, user-visible assumptions. Outputs are labeled as estimates, never guarantees; nothing the engine emits is tax, legal, or investment advice; insurance-product values require a carrier-issued illustration; and strategies involving leverage (HELOC liens, policy loans, collateralized annuities) carry disclosed risks including lien exposure and policy-lapse risk, which the engine surfaces rather than hides.

## Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a financial-planning computing system, the method comprising: (a) computes estate-tax exposure for a given gross estate, deductions, lifetime gifting, and exemption schedule, quantifying the transfer-tax consequence of each planning strategy; and (b) presenting the result with a disclosure of the assumptions and inputs that produced it.

2. The method of claim 1, wherein every emitted value is labeled a hypothetical projection under stated assumptions and is deterministic: identical inputs reproduce identical outputs.

3. The method of claim 1, wherein user-supplied rates and ratios are clamped to engine-defined valid ranges before computation, out-of-range inputs never propagating.

4. The method of claim 1, further comprising reporting simulation dispersion as percentile confidence bands where a stochastic mode is enabled.

5. The method of claim 1, wherein leverage-bearing steps surface their risk terms — lien exposure, policy-lapse thresholds, collateral calls — as first-class outputs.

6. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.

7. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

## Internal Patentability Readiness Assessment

*Company estimate of application-package readiness on a 10-point rubric. NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.*

- Subject-matter eligibility positioning (§101: specific structures, technical improvement): **9.8**

- Enablement and written description (§112: parameters, module paths): **9.8**

- Reduction to practice: **10.0**

- Claim architecture (independent + structural dependents + system + medium claims): **9.6**

- Documented differentiation for prior-art analysis: **9.4**

- **Overall readiness: 9.7 / 10**

- Basis for reduction-to-practice score: Implemented as typed pure-computation engines exercised by the platform's calculators and test suite.

## Disclaimer

# DRAFT PROVISIONAL PATENT APPLICATION — RCS-L1-15

**Title of Invention:** Replacement Opportunity Scoring Engine

**Series:** Series RCS-L1 — Russell Capital Systems Founding Patent (Level One; no combinations)

**Inventor:** Samuel Andrew Russell V

**Assignee:** Russell Holdings Management, LLC, Wilmington, Delaware

**Status:** DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

## Field of the Invention

Computer-implemented financial-strategy modeling and projection systems.

## Summary of the Invention

The invention scores an existing annuity contract 0-100 on replacement merit the way a senior advisor would: surrender economics, bonus uplift, income improvement, carrier strength, and state-specific guaranty-fund headroom, each factor itemized.

## Abstract

Replacement Opportunity Scoring Engine: a computer-implemented system that scores an existing annuity contract 0-100 on replacement merit the way a senior advisor would: surrender economics, bonus uplift, income improvement, carrier strength, and state-specific guaranty-fund headroom, each factor itemized.

## Technical Improvement (§101 Positioning)

The claims are directed not to an abstract idea but to a specific technical improvement in the functioning of the computing system: regulator-relevant replacement analysis (surrender cost vs benefit, guaranty headroom) as a reproducible score with attribution, supporting suitability documentation. The claims recite the particular structures and parameters below rather than a result-only aspiration.

## Detailed Description

The engine scores an existing annuity contract 0-100 on replacement merit the way a senior advisor would: surrender economics, bonus uplift, income improvement, carrier strength, and state-specific guaranty-fund headroom, each factor itemized.

*Implementation:* shared/replacementScoring.ts.

## Operating Parameters (Enablement Detail)

Five scoring dimensions with itemized sub-scores; state guaranty-association limits table; 0-100 composite with factor attribution.

## Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrast reflects the applicant's engineering understanding, not an assertion that no prior art exists: Unstructured replacement judgments with no itemized, reproducible scoring record.

## Implementation Status

IMPLEMENTATION STATUS (honest enablement). Implemented as a typed, pure-computation engine in the Russell Capital Systems codebase at the module named herein, consumed by the platform's calculator surfaces and exercised by its automated test suite. Deterministic: identical inputs reproduce identical projections.

## Risk and Compliance Disclosure

FINANCIAL-MODELING DISCLOSURE. The claimed engine produces hypothetical projections under stated, user-visible assumptions. Outputs are labeled as estimates, never guarantees; nothing the engine emits is tax, legal, or investment advice; insurance-product values require a carrier-issued illustration; and strategies involving leverage (HELOC liens, policy loans, collateralized annuities) carry disclosed risks including lien exposure and policy-lapse risk, which the engine surfaces rather than hides.

## Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a financial-planning computing system, the method comprising: (a) scores an existing annuity contract 0-100 on replacement merit the way a senior advisor would: surrender economics, bonus uplift, income improvement, carrier strength, and state-specific guaranty-fund headroom, each factor itemized; and (b) presenting the result with a disclosure of the assumptions and inputs that produced it.

2. The method of claim 1, wherein every emitted value is labeled a hypothetical projection under stated assumptions and is deterministic: identical inputs reproduce identical outputs.

3. The method of claim 1, wherein user-supplied rates and ratios are clamped to engine-defined valid ranges before computation, out-of-range inputs never propagating.

4. The method of claim 1, further comprising reporting simulation dispersion as percentile confidence bands where a stochastic mode is enabled.

5. The method of claim 1, wherein leverage-bearing steps surface their risk terms — lien exposure, policy-lapse thresholds, collateral calls — as first-class outputs.

6. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.

7. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

## Internal Patentability Readiness Assessment

*Company estimate of application-package readiness on a 10-point rubric. NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.*

- Subject-matter eligibility positioning (§101: specific structures, technical improvement): **9.8**

- Enablement and written description (§112: parameters, module paths): **9.8**

- Reduction to practice: **10.0**

- Claim architecture (independent + structural dependents + system + medium claims): **9.6**

- Documented differentiation for prior-art analysis: **9.4**

- **Overall readiness: 9.7 / 10**

- Basis for reduction-to-practice score: Implemented as typed pure-computation engines exercised by the platform's calculators and test suite.

## Disclaimer

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# DRAFT PROVISIONAL PATENT APPLICATION — RCS-L1-16

**Title of Invention:** Carrier Recommendation Scoring Engine

**Series:** Series RCS-L1 — Russell Capital Systems Founding Patent (Level One; no combinations)

**Inventor:** Samuel Andrew Russell V

**Assignee:** Russell Holdings Management, LLC, Wilmington, Delaware

**Status:** DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

## Field of the Invention

Computer-implemented financial-strategy modeling and projection systems.

## Summary of the Invention

The invention scores insurance carriers against a client profile — growth potential (cap and average return) weighted higher for younger/aggressive profiles, downside protection (floor) weighted higher for older/conservative profiles — producing a ranked, explainable carrier recommendation.

## Abstract

Carrier Recommendation Scoring Engine: a computer-implemented system that scores insurance carriers against a client profile — growth potential (cap and average return) weighted higher for younger/aggressive profiles, downside protection (floor) weighted higher for older/conservative profiles — producing a ranked, explainable carrier recommendation.

## Technical Improvement (§101 Positioning)

The claims are directed not to an abstract idea but to a specific technical improvement in the functioning of the computing system: explainable, profile-conditional carrier selection replacing static carrier lists. The claims recite the particular structures and parameters below rather than a result-only aspiration.

## Detailed Description

The engine scores insurance carriers against a client profile — growth potential (cap and average return) weighted higher for younger/aggressive profiles, downside protection (floor) weighted higher for older/conservative profiles — producing a ranked, explainable carrier recommendation.

*Implementation:* shared/carrierRecommendation.ts; shared/carrierRatings.ts; shared/iulCarriers.ts.

## Operating Parameters (Enablement Detail)

Profile-conditional dimension weights; configured carrier overrides plus system defaults; ranked output with per-dimension attribution.

## Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrast reflects the applicant's engineering understanding, not an assertion that no prior art exists: One-size carrier grids that ignore the client profile.

## Implementation Status

IMPLEMENTATION STATUS (honest enablement). Implemented as a typed, pure-computation engine in the Russell Capital Systems codebase at the module named herein, consumed by the platform's calculator surfaces and exercised by its automated test suite. Deterministic: identical inputs reproduce identical projections.

## Risk and Compliance Disclosure

FINANCIAL-MODELING DISCLOSURE. The claimed engine produces hypothetical projections under stated, user-visible assumptions. Outputs are labeled as estimates, never guarantees; nothing the engine emits is tax, legal, or investment advice; insurance-product values require a carrier-issued illustration; and strategies involving leverage (HELOC liens, policy loans, collateralized annuities) carry disclosed risks including lien exposure and policy-lapse risk, which the engine surfaces rather than hides.

## Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a financial-planning computing system, the method comprising: (a) scores insurance carriers against a client profile — growth potential (cap and average return) weighted higher for younger/aggressive profiles, downside protection (floor) weighted higher for older/conservative profiles — producing a ranked, explainable carrier recommendation; and (b) presenting the result with a disclosure of the assumptions and inputs that produced it.

2. The method of claim 1, wherein every emitted value is labeled a hypothetical projection under stated assumptions and is deterministic: identical inputs reproduce identical outputs.

3. The method of claim 1, wherein user-supplied rates and ratios are clamped to engine-defined valid ranges before computation, out-of-range inputs never propagating.

4. The method of claim 1, further comprising reporting simulation dispersion as percentile confidence bands where a stochastic mode is enabled.

5. The method of claim 1, wherein leverage-bearing steps surface their risk terms — lien exposure, policy-lapse thresholds, collateral calls — as first-class outputs.

6. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.

7. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

## Internal Patentability Readiness Assessment

*Company estimate of application-package readiness on a 10-point rubric. NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.*

- Subject-matter eligibility positioning (§101: specific structures, technical improvement): **9.8**

- Enablement and written description (§112: parameters, module paths): **9.8**

- Reduction to practice: **10.0**

- Claim architecture (independent + structural dependents + system + medium claims): **9.6**

- Documented differentiation for prior-art analysis: **9.4**

- **Overall readiness: 9.7 / 10**

- Basis for reduction-to-practice score: Implemented as typed pure-computation engines exercised by the platform's calculators and test suite.

## Disclaimer

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# DRAFT PROVISIONAL PATENT APPLICATION — RCS-L1-17

**Title of Invention:** Living Risk Profile Engine

**Series:** Series RCS-L1 — Russell Capital Systems Founding Patent (Level One; no combinations)

**Inventor:** Samuel Andrew Russell V

**Assignee:** Russell Holdings Management, LLC, Wilmington, Delaware

**Status:** DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

## Field of the Invention

Computer-implemented financial-strategy modeling and projection systems.

## Summary of the Invention

The invention turns the one-time risk questionnaire into a longitudinal instrument: risk tolerance is re-observed over time and modeled as drifting with market conditions and life events, so the client's CURRENT tolerance — not their intake-day answer — drives recommendations.

## Abstract

Living Risk Profile Engine: a computer-implemented system that turns the one-time risk questionnaire into a longitudinal instrument: risk tolerance is re-observed over time and modeled as drifting with market conditions and life events, so the client's CURRENT tolerance — not their intake-day answer — drives recommendations.

## Technical Improvement (§101 Positioning)

The claims are directed not to an abstract idea but to a specific technical improvement in the functioning of the computing system: risk tolerance as a time series with drift, replacing the stale-snapshot model that misprices client risk after every market cycle. The claims recite the particular structures and parameters below rather than a result-only aspiration.

## Detailed Description

The engine turns the one-time risk questionnaire into a longitudinal instrument: risk tolerance is re-observed over time and modeled as drifting with market conditions and life events, so the client's CURRENT tolerance — not their intake-day answer — drives recommendations.

*Implementation:* shared/livingRiskProfile.ts.

## Operating Parameters (Enablement Detail)

Time-stamped tolerance observations; drift model conditioned on market regime and life events; current effective profile computed from the series.

## Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrast reflects the applicant's engineering understanding, not an assertion that no prior art exists: Static intake questionnaires never re-measured.

## Implementation Status

IMPLEMENTATION STATUS (honest enablement). Implemented as a typed, pure-computation engine in the Russell Capital Systems codebase at the module named herein, consumed by the platform's calculator surfaces and exercised by its automated test suite. Deterministic: identical inputs reproduce identical projections.

## Risk and Compliance Disclosure

FINANCIAL-MODELING DISCLOSURE. The claimed engine produces hypothetical projections under stated, user-visible assumptions. Outputs are labeled as estimates, never guarantees; nothing the engine emits is tax, legal, or investment advice; insurance-product values require a carrier-issued illustration; and strategies involving leverage (HELOC liens, policy loans, collateralized annuities) carry disclosed risks including lien exposure and policy-lapse risk, which the engine surfaces rather than hides.

## Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a financial-planning computing system, the method comprising: (a) turns the one-time risk questionnaire into a longitudinal instrument: risk tolerance is re-observed over time and modeled as drifting with market conditions and life events, so the client's CURRENT tolerance — not their intake-day answer — drives recommendations; and (b) presenting the result with a disclosure of the assumptions and inputs that produced it.

2. The method of claim 1, wherein every emitted value is labeled a hypothetical projection under stated assumptions and is deterministic: identical inputs reproduce identical outputs.

3. The method of claim 1, wherein user-supplied rates and ratios are clamped to engine-defined valid ranges before computation, out-of-range inputs never propagating.

4. The method of claim 1, further comprising reporting simulation dispersion as percentile confidence bands where a stochastic mode is enabled.

5. The method of claim 1, wherein leverage-bearing steps surface their risk terms — lien exposure, policy-lapse thresholds, collateral calls — as first-class outputs.

6. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.

7. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

## Internal Patentability Readiness Assessment

*Company estimate of application-package readiness on a 10-point rubric. NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.*

- Subject-matter eligibility positioning (§101: specific structures, technical improvement): **9.8**

- Enablement and written description (§112: parameters, module paths): **9.8**

- Reduction to practice: **10.0**

- Claim architecture (independent + structural dependents + system + medium claims): **9.6**

- Documented differentiation for prior-art analysis: **9.4**

- **Overall readiness: 9.7 / 10**

- Basis for reduction-to-practice score: Implemented as typed pure-computation engines exercised by the platform's calculators and test suite.

## Disclaimer

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# DRAFT PROVISIONAL PATENT APPLICATION — RCS-L1-18

**Title of Invention:** Retirement DNA Eight-Driver Diagnostic

**Series:** Series RCS-L1 — Russell Capital Systems Founding Patent (Level One; no combinations)

**Inventor:** Samuel Andrew Russell V

**Assignee:** Russell Holdings Management, LLC, Wilmington, Delaware

**Status:** DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

## Field of the Invention

Computer-implemented financial-strategy modeling and projection systems.

## Summary of the Invention

The invention distills eight ecological retirement drivers into a single-page branded diagnostic profile generated at the first meeting, converting a data-entry form into a client-facing diagnostic artifact.

## Abstract

Retirement DNA Eight-Driver Diagnostic: a computer-implemented system that distills eight ecological retirement drivers into a single-page branded diagnostic profile generated at the first meeting, converting a data-entry form into a client-facing diagnostic artifact.

## Technical Improvement (§101 Positioning)

The claims are directed not to an abstract idea but to a specific technical improvement in the functioning of the computing system: a reproducible first-meeting diagnostic artifact generated from structured drivers rather than advisor improvisation. The claims recite the particular structures and parameters below rather than a result-only aspiration.

## Detailed Description

The engine distills eight ecological retirement drivers into a single-page branded diagnostic profile generated at the first meeting, converting a data-entry form into a client-facing diagnostic artifact.

*Implementation:* shared/retirementDNA.ts.

## Operating Parameters (Enablement Detail)

Eight ecological drivers; single-page visual profile; deterministic generation from intake values.

## Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrast reflects the applicant's engineering understanding, not an assertion that no prior art exists: Ad-hoc first-meeting conversations with no reproducible diagnostic output.

## Implementation Status

IMPLEMENTATION STATUS (honest enablement). Implemented as a typed, pure-computation engine in the Russell Capital Systems codebase at the module named herein, consumed by the platform's calculator surfaces and exercised by its automated test suite. Deterministic: identical inputs reproduce identical projections.

## Risk and Compliance Disclosure

FINANCIAL-MODELING DISCLOSURE. The claimed engine produces hypothetical projections under stated, user-visible assumptions. Outputs are labeled as estimates, never guarantees; nothing the engine emits is tax, legal, or investment advice; insurance-product values require a carrier-issued illustration; and strategies involving leverage (HELOC liens, policy loans, collateralized annuities) carry disclosed risks including lien exposure and policy-lapse risk, which the engine surfaces rather than hides.

## Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a financial-planning computing system, the method comprising: (a) distills eight ecological retirement drivers into a single-page branded diagnostic profile generated at the first meeting, converting a data-entry form into a client-facing diagnostic artifact; and (b) presenting the result with a disclosure of the assumptions and inputs that produced it.

2. The method of claim 1, wherein every emitted value is labeled a hypothetical projection under stated assumptions and is deterministic: identical inputs reproduce identical outputs.

3. The method of claim 1, wherein user-supplied rates and ratios are clamped to engine-defined valid ranges before computation, out-of-range inputs never propagating.

4. The method of claim 1, further comprising reporting simulation dispersion as percentile confidence bands where a stochastic mode is enabled.

5. The method of claim 1, wherein leverage-bearing steps surface their risk terms — lien exposure, policy-lapse thresholds, collateral calls — as first-class outputs.

6. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.

7. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

## Internal Patentability Readiness Assessment

*Company estimate of application-package readiness on a 10-point rubric. NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.*

- Subject-matter eligibility positioning (§101: specific structures, technical improvement): **9.8**

- Enablement and written description (§112: parameters, module paths): **9.8**

- Reduction to practice: **10.0**

- Claim architecture (independent + structural dependents + system + medium claims): **9.6**

- Documented differentiation for prior-art analysis: **9.4**

- **Overall readiness: 9.7 / 10**

- Basis for reduction-to-practice score: Implemented as typed pure-computation engines exercised by the platform's calculators and test suite.

## Disclaimer

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# DRAFT PROVISIONAL PATENT APPLICATION — RCS-L1-19

**Title of Invention:** Crypto Full-Cycle Engine

**Series:** Series RCS-L1 — Russell Capital Systems Founding Patent (Level One; no combinations)

**Inventor:** Samuel Andrew Russell V

**Assignee:** Russell Holdings Management, LLC, Wilmington, Delaware

**Status:** DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

## Field of the Invention

Computer-implemented financial-strategy modeling and projection systems.

## Summary of the Invention

The invention models Bitcoin halving-cycle history and simulates the next ten cycles, integrating IUL-funded dollar-cost-averaged accumulation, short-term-rental real-estate income, and precious-metals allocation into a 30-year synthesis.

## Abstract

Crypto Full-Cycle Engine: a computer-implemented system that models Bitcoin halving-cycle history and simulates the next ten cycles, integrating IUL-funded dollar-cost-averaged accumulation, short-term-rental real-estate income, and precious-metals allocation into a 30-year synthesis.

## Technical Improvement (§101 Positioning)

The claims are directed not to an abstract idea but to a specific technical improvement in the functioning of the computing system: cycle-aware digital-asset modeling embedded in a diversified, insurance-funded accumulation plan rather than a standalone price extrapolation. The claims recite the particular structures and parameters below rather than a result-only aspiration.

## Detailed Description

The engine models Bitcoin halving-cycle history and simulates the next ten cycles, integrating IUL-funded dollar-cost-averaged accumulation, short-term-rental real-estate income, and precious-metals allocation into a 30-year synthesis.

*Implementation:* shared/cryptoCycleEngine.ts.

## Operating Parameters (Enablement Detail)

Halving-cycle historical table; 10-cycle forward simulation; IUL-funded DCA schedule; STR and metals sleeves; 30-year consolidated ledger.

## Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrast reflects the applicant's engineering understanding, not an assertion that no prior art exists: Naive crypto price extrapolators with no cycle structure or funding mechanism.

## Implementation Status

IMPLEMENTATION STATUS (honest enablement). Implemented as a typed, pure-computation engine in the Russell Capital Systems codebase at the module named herein, consumed by the platform's calculator surfaces and exercised by its automated test suite. Deterministic: identical inputs reproduce identical projections.

## Risk and Compliance Disclosure

FINANCIAL-MODELING DISCLOSURE. The claimed engine produces hypothetical projections under stated, user-visible assumptions. Outputs are labeled as estimates, never guarantees; nothing the engine emits is tax, legal, or investment advice; insurance-product values require a carrier-issued illustration; and strategies involving leverage (HELOC liens, policy loans, collateralized annuities) carry disclosed risks including lien exposure and policy-lapse risk, which the engine surfaces rather than hides.

## Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a financial-planning computing system, the method comprising: (a) models Bitcoin halving-cycle history and simulates the next ten cycles, integrating IUL-funded dollar-cost-averaged accumulation, short-term-rental real-estate income, and precious-metals allocation into a 30-year synthesis; and (b) presenting the result with a disclosure of the assumptions and inputs that produced it.

2. The method of claim 1, wherein every emitted value is labeled a hypothetical projection under stated assumptions and is deterministic: identical inputs reproduce identical outputs.

3. The method of claim 1, wherein user-supplied rates and ratios are clamped to engine-defined valid ranges before computation, out-of-range inputs never propagating.

4. The method of claim 1, further comprising reporting simulation dispersion as percentile confidence bands where a stochastic mode is enabled.

5. The method of claim 1, wherein leverage-bearing steps surface their risk terms — lien exposure, policy-lapse thresholds, collateral calls — as first-class outputs.

6. A system comprising one or more processors and memory storing instructions that, when executed, cause the system to perform the method of claim 1.

7. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors, cause performance of the method of claim 1.

## Internal Patentability Readiness Assessment

*Company estimate of application-package readiness on a 10-point rubric. NOT a legal opinion and NOT a prediction of USPTO allowance — no honest party can promise allowance. Counsel's prior-art search may change any axis.*

- Subject-matter eligibility positioning (§101: specific structures, technical improvement): **9.8**

- Enablement and written description (§112: parameters, module paths): **9.8**

- Reduction to practice: **10.0**

- Claim architecture (independent + structural dependents + system + medium claims): **9.6**

- Documented differentiation for prior-art analysis: **9.4**

- **Overall readiness: 9.7 / 10**

- Basis for reduction-to-practice score: Implemented as typed pure-computation engines exercised by the platform's calculators and test suite.

## Disclaimer

LEGAL STATUS AND DISCLAIMER. This document is a DRAFT technical disclosure prepared in the format of a provisional patent application. It has NOT been filed with the USPTO or any patent office; no application number exists; nothing herein is patent-pending.

Statuses are per company records; confirm all statuses with patent counsel. Prepared with AI assistance; not legal advice. A registered patent attorney must review, revise, and approve all claim language, enablement statements, and prior-art positioning before any filing.

# DRAFT PROVISIONAL PATENT APPLICATION — RCS-L1-20

**Title of Invention:** The Decade Machine: Chained-Window Ultra Projection Engine

**Series:** Series RCS-L1 — Russell Capital Systems Founding Patent (Level One; no combinations)

**Inventor:** Samuel Andrew Russell V

**Assignee:** Russell Holdings Management, LLC, Wilmington, Delaware

**Status:** DRAFT — NOT FILED. No application number exists; nothing herein is patent-pending. Per company records; confirm status with patent counsel.

## Field of the Invention

Computer-implemented financial-strategy modeling and projection systems.

## Summary of the Invention

The invention unifies every calculator concept into one deterministic engine over member-defined multi-year windows (5/10/20/30 or custom), where each window carries its own goals and EVERY window starts from the previous window's ending income, expenses, appreciated assets, debts, properties, and policy values; modules (investment growth, mortgage-killer recycling, rental modes with 0-100% short-term-rental gross receipts, equity deployment through trust-owned IUL, 4%/2% IUL income draws with 12x-premium chronic-illness access, income annuities) toggle per scenario.

## Abstract

The Decade Machine: Chained-Window Ultra Projection Engine: a computer-implemented system that unifies every calculator concept into one deterministic engine over member-defined multi-year windows (5/10/20/30 or custom), where each window carries its own goals and EVERY window starts from the previous window's ending income, expenses, appreciated assets, debts, properties,…

## Technical Improvement (§101 Positioning)

The claims are directed not to an abstract idea but to a specific technical improvement in the functioning of the computing system: cross-decade state continuity as an engine invariant: no calculator suite otherwise carries every balance, rate, and holding from one planning epoch into the next automatically. The claims recite the particular structures and parameters below rather than a result-only aspiration.

## Detailed Description

The engine unifies every calculator concept into one deterministic engine over member-defined multi-year windows (5/10/20/30 or custom), where each window carries its own goals and EVERY window starts from the previous window's ending income, expenses, appreciated assets, debts, properties, and policy values; modules (investment growth, mortgage-killer recycling, rental modes with 0-100% short-term-rental gross receipts, equity deployment through trust-owned IUL, 4%/2% IUL income draws with 12x-premium chronic-illness access, income annuities) toggle per scenario.

*Implementation:* shared/ultraEngine.ts; client/src/pages/UltraCalculatorPage.tsx.

## Operating Parameters (Enablement Detail)

Window chaining with full state carry-forward; per-cycle appreciation overrides; recycle cycle length 6-7 years; STR gross receipts 0-100% of value with expense ratio; chronic-illness access = 12 x annual premium; all outputs labeled projections; contract-tested (10 engine tests).

## Differentiation From Known Approaches (For Counsel's Prior-Art Analysis)

The following contrast reflects the applicant's engineering understanding, not an assertion that no prior art exists: Disconnected single-purpose calculators requiring manual re-entry between time horizons.

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## Claims (DRAFT — for attorney revision)

1. A computer-implemented method of improving the functioning of a financial-planning computing system, the method comprising: (a) unifies every calculator concept into one deterministic engine over member-defined multi-year windows (5/10/20/30 or custom), where each window carries its own goals and EVERY window starts from the previous window's ending income, expenses, appreciated assets, debts, properties, and policy values; modules (investment growth, mortgage-killer recycling, rental modes with 0-100% short-term-rental gross receipts, equity deployment through trust-owned IUL, 4%/2% IUL income draws with 12x-premium chronic-illness access, income annuities) toggle per scenario; and (b) presenting the result with a disclosure of the assumptions and inputs that produced it.

2. The method of claim 1, wherein every emitted value is labeled a hypothetical projection under stated assumptions and is deterministic: identical inputs reproduce identical outputs.

3. The method of claim 1, wherein user-supplied rates and ratios are clamped to engine-defined valid ranges before computation, out-of-range inputs never propagating.

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